

# Queue Cure '26

## Hackathon Thought Process Sheet

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**Project:** Real-Time Hospital Queue Management System

### 1. PROBLEM STATEMENT

Many clinics still use manual token systems. Patients often do not know their position in the queue or estimated waiting time. Receptionists manually manage patient flow, which can cause confusion and delays.

### 2. PROPOSED SOLUTION

Queue Cure is a real-time hospital queue management system that digitizes the patient queue process. It allows receptionists to manage patient tokens while providing patients with live updates about their waiting status.

### 3. KEY FEATURES

- Add patients with automatic token generation.
- Call the next patient in sequence.
- Real-Time Queue Synchronization.
- Current Token Display.
- People Ahead Calculation.
- Estimated Waiting Time Calculation.

### 4. TECHNOLOGY STACK

|                                |                            |
|--------------------------------|----------------------------|
| <b>Frontend:</b>               | HTML5, CSS3, JavaScript    |
| <b>Backend &amp; Database:</b> | Firebase Realtime Database |
| <b>Version Control:</b>        | GitHub                     |

### 5. CORE WORKFLOW

- 1 Receptionist enters patient name.
- 2 System generates a unique token.
- 3 Patient data is stored in Firebase Realtime Database.
- 4 Receptionist clicks "Call Next".
- 5 Current token updates in Firebase.
- 6 Patient screen automatically receives updates.
- 7 Queue information is refreshed without page reload.

## 6. CONCURRENCY HANDLING

- Multiple users can view the queue simultaneously.
- Firebase Realtime Database instantly syncs updates across connected screens.
- FIFO (First In First Out) logic ensures patients are served in order.
- Token numbers are generated sequentially to avoid duplication.

## 7. EDGE CASES HANDLED

- Empty patient name validation.
- Empty queue handling.
- Preventing invalid token calculations.
- Automatic update when queue becomes empty.

## 8. FUTURE IMPROVEMENTS

1. Doctor Dashboard
2. Online Appointment Booking
3. SMS Notifications
4. Multi-Clinic Support
5. Patient History Records

## 9. CONCLUSION

Queue Cure provides a simple and effective solution for managing patient queues in clinics. By using Firebase Realtime Database, the system enables real-time synchronization between receptionist and patient screens, improving transparency and reducing waiting confusion.